

● EASY

● ALGEBRA

Linear functions

04

10 questions · ~15 min

Q1

ALGEBRA

The function f is defined by $f(x) = 3x - 8$. What is the value of $f7$?

- A. 29
- B. 13
- C. -5
- D. -29

A student council group is selling school posters for a fundraiser. They use the function $px = 5x - 220$ to determine their profit px , in dollars, for selling x school posters. In order to earn a profit of \$ 900, how many school posters must they sell?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ALGEBRA

A veterinarian recommends that each day a certain rabbit should eat 25 calories per pound of the rabbit's weight, plus an additional 11 calories. Which equation represents this situation, where c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit's weight is x pounds?

- A. $c = 25x$
- B. $c = 36x$
- C. $c = 11x + 25$
- D. $c = 25x + 11$

Q4

ALGEBRA

$$f(x) = x + b$$

For the linear function f , b is a constant. When $x = 0$, $f(x) = 30$. What is the value of b ?

- A. -30
- B. $-\frac{1}{30}$
- C. $\frac{1}{30}$
- D. 30

Q5

GRID-IN ALGEBRA

$$f(x) = 7x + 1$$

The function gives the total number of people on a company retreat with x managers. What is the total number of people on a company retreat with 7 managers?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$d = 16t$$

The given equation represents the distance d , in inches, where t represents the number of seconds since an object started moving. Which of the following is the best interpretation of 16 in this context?

- A. The object moved a total of 16 inches.
- B. The object moved a total of $16t$ inches.
- C. The object is moving at a rate of 16 inches per second.
- D. The object is moving at a rate of $\frac{1}{16}$ inches per second.

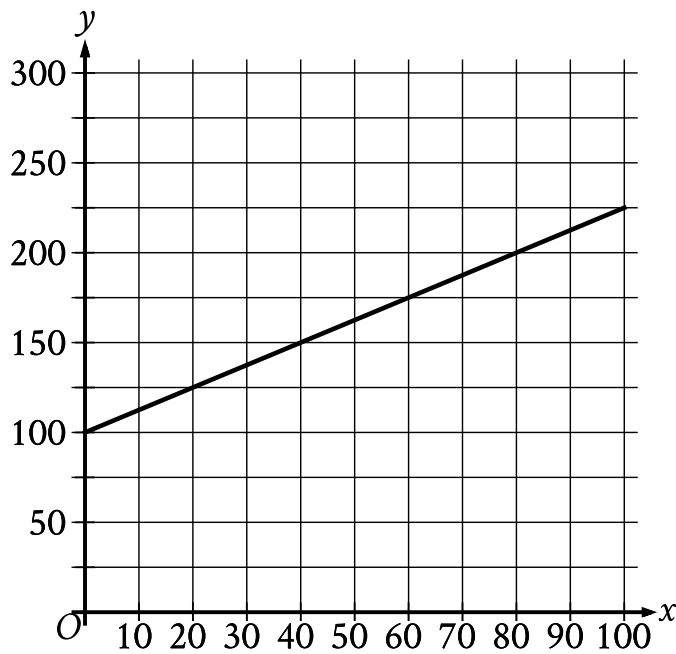
In the linear function h , $h(0) = 41$ and $h(1) = 40$. Which equation defines h ?

- A. $h(x) = -x + 41$
- B. $h(x) = -x$
- C. $h(x) = -41x$
- D. $h(x) = -41$

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d , in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker?

- A. 30
- B. 32
- C. 60
- D. 90

The cost y , in dollars, for a manufacturer to make x rings is represented by the line shown.



- The line slants gradually up from left to right.
- The line passes through the following points:
 - (0 comma 100)
 - (60 comma 175)
 - (100 comma 225)

What is the cost, in dollars, for the manufacturer to make 60 rings?

- A. 100
- B. 125
- C. 175

D. 225

Q10

ALGEBRA

The function f is defined by $fx = 5x + 8$. For what value of x does $fx = 58$?

A. 10

B. 13

C. 50

D. 298